MARK SCHEME NOTES

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

Types of mark

M Method marks, awarded for a valid method applied to the problem.

A Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. For accuracy marks to be given, the associated Method mark must be earned or implied.

B Mark for a correct result or statement independent of Method marks.

When a part of a question has two or more 'method' steps, the M marks are in principle independent unless the scheme specifically says otherwise; and similarly where there are several B marks allocated. The notation 'dep' is used to indicate that a particular M or B mark is dependent on an earlier mark in the scheme.

Abbreviations

awrt	answers which round to
cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
nfww	not from wrong working
oe	or equivalent
rot	rounded or truncated
SC	Special Case
soi	seen or implied

Question	Answer	Marks	Partial Marks
1(a)	radius	**1**	
1(b)	chord	**1**	
2	1 2 4 8 16 32	**2**	**B1** for 4 correct and no errors or 6 correct and one extra
3	< = <	**2**	**B1** for 2 correct
4(a)	–2, 3	**1**	
4(b)	correct ruled line $y = x$	**1**	
4(c)	correct ruled line $y = -x + 1$	**1**	**FT** *their* straight $y = x$
5(a)	$\frac{5}{8}$ or 0.625	**1**	

Question	Answer	Marks	Partial Marks
5(b)	−3	**1**	
6(a)	1 or 81 accept 0	**1**	
6(b)	1	**1**	
6(c)	59	**1**	
7	20	**1**	
8	650 3250	**2**	**B1** for each
9(a)	trapezium	**1**	
9(b)	rhombus	**1**	
10	3.06 or 3.057…	**3**	**M1** for $(1 \times 7) + (2 \times 5) + (3 \times 8) + (4 \times 9) + (5 \times 6)$ **M1dep** for $their \sum fx \div (7+5+8+9+6)$
11	102	**2**	**B1** for 108 or – 6 or **M1** for $3 \times 6^2 + 2 \times -3$ oe
12(a)	64 136 160	**2**	**B1** for 1 correct or **M1** for $\frac{360}{90} \times k$ (k = 1, 16, 34 or 40) oe
12(b)	Accurate pie chart	**2**	**B2FT** if *their* angles add up to 360° or **B1FT** for one correct sector or one correct FT sector (may or may not add up to 360°)
13	32 800 cao	**2**	**B1** for 32 768 or *their* answer of more than 3 figures correctly rounded to 3 sf
14(a)	40	**2**	**B1** for 8 or $22\frac{1}{2}$ or $17\frac{1}{2}$ if 0 scored **SC1** for 45 as final answer
14(b)	608	**1**	**FT** *their* **(a)** × 15.20 correct to at least 3sf
14(c)	22.8[0]	**1**	
14(d)	400	**1**	
15	Correct point marked	**2**	**B1** for *B* on bearing 140° from *A* or **B1** for *B* 5.5cm from *A*